WILLIAM FAURIAT

Senior Data Scientist & AI/ML Engineer

+ 33(0)7.69.-.-.- | wfauriat@gmail.com

Profile & Web-CV @ wfauriat.github.io | *Developments & Projects @ github.com/wfauriat*

Summary

10+ years of experience in data science across diverse industrial and scientific domains (automotive, maintenance, high-energy physics). Research engineer specialized in Uncertainty Quantification (UQ), Machine Learning (ML), and decision-making under uncertainty. Today, I want to dedicate my expertise to building data-driven solutions: production-grade APIs, robust ML pipelines, and AI/LLM-powered automation.

Experience

Data Scientist 2021–present

CEA (French Alternative Energies and Atomic Energy Commission) - Military Applications Division

- Audit of high-power tomography algorithms and uncertainty quantification
- Development of analysis tools (APIs) and visualization aids (GUIs) for data exploitation
- Bayesian inference-based calibration of simulation codes from experimental data
- Robust optimization and ML for the design of inertial confinement fusion experiments
- Training coordinator (internal sessions, 20 people): uncertainty handling in simulations

Post-doctoral Researcher 2018–2020

CentraleSupélec - Industrial Engineering Laboratory

- Research on optimal information collection – Bayesian statistical decision theory
- Application to inspection policy design for condition-based maintenance
- Over 100 hours of teaching in systems reliability and industrial engineering – Master's level

Sabbatical Year 2017

Travel across Asia, Oceania, and South America

Consultant Engineer 2016

ALTEN for PSA (now Stellantis)

- R&D Engineer: design and validation of vehicle suspension systems

Doctoral Research 2013–2016

Renault – Université Clermont Auvergne (Industrial PhD – CIFRE)

- Probabilistic modeling of road-induced loads – validation in vibration fatigue
- Exploitation of vehicle measurements for road profile estimation

Education

Full-Stack Developer Certification 2024

Codecademy (online) 150+ hours of courses, projects and exams: React, Node.js, REST APIs, SQL, Git, CI/CD

PhD in Mechanical Engineering 2016

Université Clermont Auvergne

Engineering Degree in Mechanics 2012

IFMA – Institut Français de Mécanique Avancée (now SIGMA Clermont)

Skills & Expertise

Expertise

- Uncertainty Quantification (UQ)
- Statistical Learning / Machine Learning (ML)
- Bayesian Inference
- Full-Stack Development
- API design – GUI design
- LLM-assisted co-development – AI agents

Tech Stack

- **Languages:** Python (expert), JavaScript (intermediate), SQL, R, C++ (familiar)
- **Frontend (Web & desktop):** HTML/CSS, React, PyQt5
- **Backend:** Flask - FastAPI (Python), Node.js, Express (JS)
- **Data/ML:** NumPy, SciPy, pandas, scikit-learn, PyTorch
- **Tools:** Linux, Git, Docker, VSCode

Selected Publications

Fauriat, W. & Zio, E. (2020). Optimization of an aperiodic sequential inspection and condition-based maintenance policy driven by value of information. *Reliability Engineering & System Safety*

Fauriat, W., Mattrand, C., Gayton, N., Beakou, A., & Cembrzynski, T. (2016). Estimation of road profile variability from measured vehicle responses. *Vehicle System Dynamics*

Fauriat, W. & Gayton, N. (2014). AK-SYS: An adaptation of the AK-MCS method for system reliability. *Reliability Engineering & System Safety*

Developments – Projects (see @ Github)

Bayesian Inference Learning & Experimentation GUI: Full-Stack application: Bayesian inference core in pure Python (numpy) – Two frontend versions, one in PyQt5 (desktop), one in React (Web). Backend (REST API) in Flask + Python with ML model comparison (sklearn) | deployed at bi-webapp.onrender.com (some waiting at cold start)

Other (training or ongoing projects): PyQt GUI (UQ introduction): pure desktop | "Argument Mapping" Web App: pure web (React, React Flow) | Electricity demand forecasting: end-to-end ML pipeline (external API call, FastAPI backend, MLflow experiment tracking, Prefect orchestration, Streamlit dashboard, Docker containerization)